

Homework Solutions — Question 11.6

MANOVA with Small Sample: Problem Identification and Rectification

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$n = 6$

Total observations
(3 per group)

$p = 4$

Dependent variables
(Y_1, Y_2, Y_3, Y_4)

$g = 2$

Groups

Rank $W = 3$

Within-group matrix
rank-deficient

Question 11.6 Data, Group Means, and SSCP Matrices

11.6.1 Raw Data

Obs.	Y ₁	Y ₂	Y ₃	Y ₄	Group
1	7	3	1	2	1
2	8	2	3	3	1
3	9	1	5	5	1
4	9	5	3	4	2
5	10	4	5	5	2
6	11	8	7	6	2

11.6.2 Group Means and Grand Mean

	$\bar{Y}_1$	$\bar{Y}_2$	$\bar{Y}_3$	$\bar{Y}_4$
Group 1	8.0000	2.0000	3.0000	3.3333
Group 2	10.0000	5.6667	5.0000	5.0000
Grand Mean	9.0000	3.8333	4.0000	4.1667

11.6.3 Within-Group Matrix W and Between-Group Matrix B

Within-Group Matrix W:

	Y ₁	Y ₂	Y ₃	Y ₄
Y ₁	4.00	1.00	8.00	5.00
Y ₂	1.00	10.67	2.00	0.00
Y ₃	8.00	2.00	16.00	10.00
Y ₄	5.00	0.00	10.00	6.67

Between-Group Matrix B:

	Y ₁	Y ₂	Y ₃	
Y ₁	6.00	11.00	6.00	
Y ₂	11.00	20.17	11.00	
Y ₃	6.00	11.00	6.00	
Y ₄	5.00	9.17	5.00	

Question 11.6 MANOVA Results and Problem Identification

11.6.4 MANOVA Test Results (Full Model, p = 4)

Quantity	Value	Formula	df
Rank of W (within-group matrix)	3 ← singular!	rank(W)	should = 4
det(W)	0.000000	det(E)	—
det(T)	66.666667	det(E+H)	—
Wilks' Lambda Λ	0.000000 ← degenerate	det(W)/det(T)	—
df ₂ = n – g – p + 1	1 (should be > 0)	exact formula	—
F-approximation	INVALID — df ₂ = 1	F(p, df ₂)	p=4, df ₂ =1

11.6.5 Identification of the Problem

The fundamental problem is that the sample size is too small relative to the number of dependent variables.

MANOVA requires: n – g > p (within-group degrees of freedom must exceed the number of DVs)

Here: n – g = 6 – 2 = 4, p = 4 ⇒ 4 > 4 is FALSE

Consequences:

11.6.6 Eigenvalues of W (Confirming Singularity)

- (1) The within-group Wilks' Lambda = 0.000000. Its determinant = 0.
- (2) Wilks' Lambda = det(W)/det(T) = 0 — a degenerate result, not a true test statistic.
- (3) The denominator degrees of freedom df₂ = n – g – p + 1 = 1 makes the F test essentially powerless.
- (4) Statistical conclusions drawn from this analysis are unreliable and not generalizable.

Eigenvalue	Value	Non-zero?	Interpretation
λ ₁	26.588543	Yes	Non-zero — valid variance dimension
λ ₂	10.546676	Yes	Non-zero — valid variance dimension
λ ₃	0.198115	Yes	Near-zero — minimal variance
λ ₄	0.000000	— 3 — No	Exactly zero — SINGULAR (confirms rank deficiency)

Question 11.6 How to Rectify the Problem

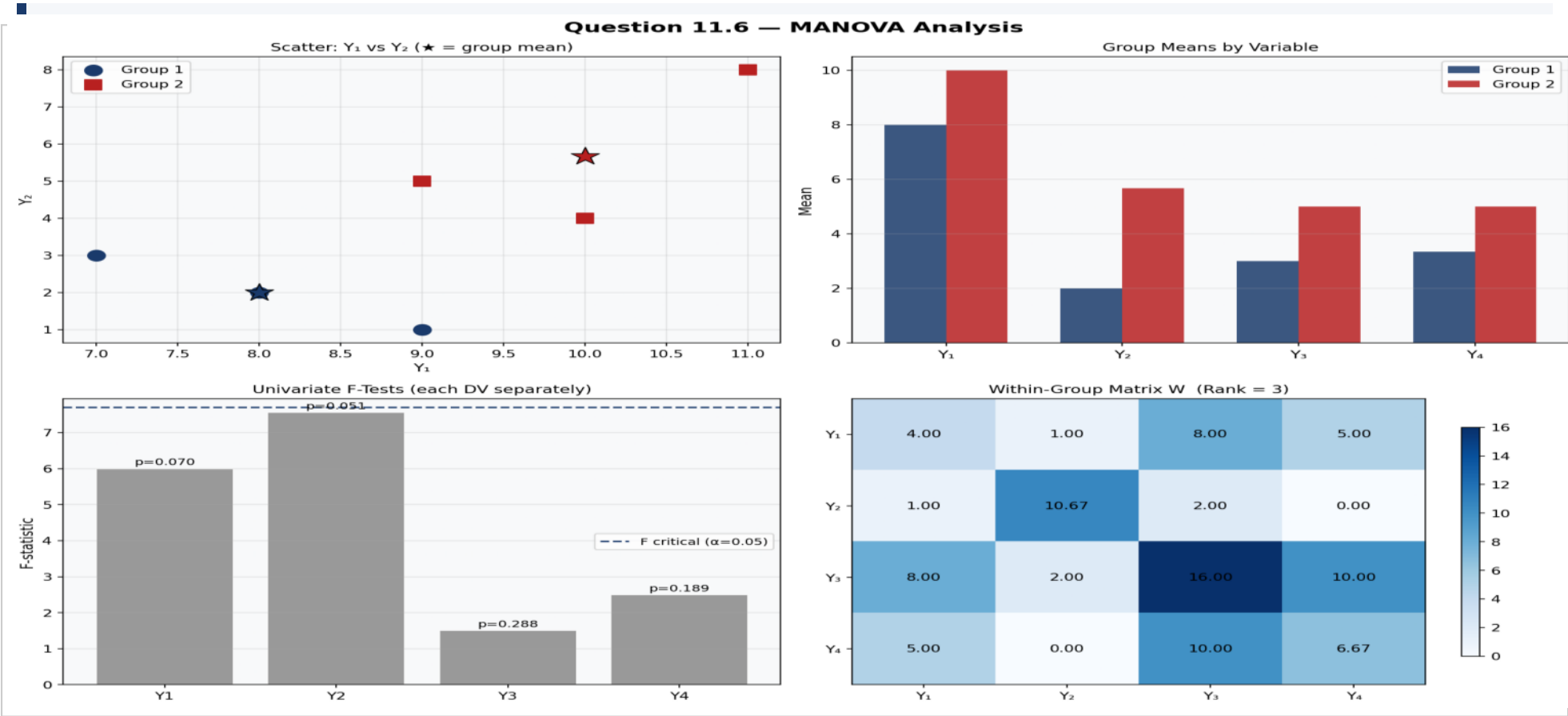
A	Option A: Increase Sample Size 6.7 Recommended Remedies Collect more observations so that $n - g > p$. The minimum requirement is $n > g + p = 6$, but a practical rule of thumb is at least $10 \times p$ observations per group (i.e., 40+ per group, 80+ total). This is the most statistically principled remedy.
B	Option B: Reduce Number of DVs Use a subset of $p' < p$ dependent variables such that $n - g > p'$. For example, using only Y_1 and Y_2 gives $p' = 2$, and $n - g = 4 > 2$ (ok). Variable selection should be guided by theory, not by which variables give significant results (to avoid data dredging).
C	Option C: Dimension Reduction via PCA Apply Principal Component Analysis to Y_1-Y_4 first, retaining fewer components (e.g., 2 components) that explain most of the variance. Then run MANOVA on the component scores. This preserves information while satisfying $n - g > p$.
D	Option D: Univariate ANOVAs Run separate one-way ANOVAs for each dependent variable. While this approach ignores the multivariate structure and inflates Type I error, it remains valid for each DV individually when MANOVA is infeasible. Apply Bonferroni correction for multiple tests.

11.6.8 Remedy B Applied: MANOVA with p=2 (Y1, Y2 only)

Quantity	Value	df ₁	df ₂	p-value
Wilks' Lambda (Y ₁ ,Y ₂)	0.253550	2	3	0.1277

$F(2, 3) = 4.4160$, $p = 0.1277$. With only $p=2$ DVs the test is now technically valid ($df_2 = 3 > 0$), but the result is not significant — reflecting the very small sample size and low power.

Question 11.6 Figures and Univariate ANOVA Summary



ht) Heatmap of within-group matrix W with rank annotation.